B. M. S. COLLEGE OF ENGINEERING, BANGALORE – 560019
DEPARTMENT OF MATHEMATICS

TRANSFORM CALCULUS, FOURIER SERIES AND NUMERICAL TECHNIQUES

UNIT-3: FOURIER TRANSFORMS (Course Code:23MA3BSTFN)

Formulae :-

The Fourier transform of $f(x)$ is given by $F(s)=\int_{-\infty}^{\infty} f(x)e^{isx}\,dx$

The Inverse Fourier transform of $F(s)$ is given by $f(x)=\frac{1}{2\pi}\int_{-\infty}^{\infty} F(s)\,e^{-isx}ds$

The Fourier Sine Transform of $f(x)$ is given by $F_s(s)=\int_0^{\infty} f(x)\sin sx\,dx$

The Fourier Cosine Transform of $f(x)$ is given by $F_c(s)=\int_0^{\infty} f(x)\cos sx\,dx$

The Inverse Fourier Sine transform of $F_s(s)$ is given by $f(x)=\frac{2}{\pi}\int_0^{\infty} F_s(s)\sin sx\,ds$

The Inverse Fourier Cosine transform of $F_c(s)$ is given by $f(x)=\frac{2}{\pi}\int_0^{\infty} F_c(s)\cos sx\,ds$

Infinite Fourier transforms

1. Find the Fourier transform of the following functions.

   a. $f(x)=\begin{cases}\sin x & 0<x<\pi\\ 0 & otherwise\end{cases}$. **Ans:** $\frac{1+e^{is\pi}}{1-s^2}$

   b. $f(x)=\begin{cases}\frac{1}{2a} & \text{if } |x|\le a\\ 0 & \text{if } |x|>a\end{cases}$. **Ans:** $\frac{\sin(as)}{as}$

   c. $f(x)=\begin{cases}x & \text{for } |x|\le a\\ 0 & \text{for } |x|>a\end{cases}$. **Ans:** $\frac{2i}{s^2}\left[\sin as-as\cos(as)\right]$

   d. $f(x)=\begin{cases}0 & -\infty<x<a\\ x & a\le x\le b\\ 0 & x>b\end{cases}$. **Ans:** $\frac{1}{s}\left(ae^{isa}-be^{isb}\right)+\frac{1}{s^2}\left(e^{isb}-e^{isa}\right)$

   e. $f(x)=\begin{cases}\cos x & 0<x<1\\ 0 & \text{otherwise}\end{cases}$. **Ans:** $\frac{1}{1-s^2}\left[e^{is}\left\{is\cos 1+\sin 1\right\}-is\right]$

   or $\frac{1}{2}\left[\left(\frac{\sin(s+1)}{s+1}+\frac{\sin(s-1)}{s-1}\right)+i\left(\frac{1-\cos(s+1)}{s+1}+\frac{1-\cos(s-1)}{s-1}\right)\right]$.

   f. $f(x)=e^{-|x|}$. **Ans:** $\frac{2}{1+s^2}$

2. Find the Fourier transform of $e^{-a^2x^2}$, $a>0$.

3. Find the Fourier transform of $f(x)=\begin{cases}1 & |x|\le 1\\ 0 & |x|>1\end{cases}$. **Ans:** $F(s)=\frac{2\sin s}{s}$

4. Find the Fourier transform of $f(x)=\begin{cases}a^2-x^2 & |x|\le a\\ 0 & |x|>a\end{cases}$

Ans: $F(s)=\frac{-4}{s^3}\left[as\cos as-\sin as\right]$

5. Find the Fourier transform of $f(x)=\begin{cases}1-|x| & |x|\le 1\\ 0 & |x|>1\end{cases}$ **Ans:** $F\left[f(x)\right]=\frac{2}{s^2}\left[1-\cos(s)\right]$

<u>Infinite Inverse Fourier transforms</u>

Find the inverse Fourier transform of

i) e^{-s^2} **Ans:** $\frac{e^{-x^2/4}}{2\sqrt{\pi}}$ ii) $e^{-|s|a}$ **Ans:** $\frac{a}{\pi\left[a^2+x^2\right]}$ iii) $F(s)=\begin{cases}s & \text{for } |s|\le a\\ 0 & \text{for } |s|>a\end{cases}$

iv) $\boldsymbol{F}(s)=\begin{cases}\cos s & 0<s<1\\ 0 & \text{otherwise}\end{cases}$

<u>Infinite Fourier cosine transforms</u>

1. Find the Fourier cosine transforms of the following functions.

i) $f(x)=e^{-x^2}$ **Ans**: $\frac{\sqrt{\pi}}{2}e^{-(s^2/4)}$

ii) $f(x)=e^{-ax}\cos ax$ **Ans:** $\frac{a\left(s^2+2a^2\right)}{s^4+4a^4}$

iii) $f(x)=\begin{cases}x & 0<x<1\\ 2-x & 1<x<2\\ 0 & \text{otherwise}\end{cases}$ **Ans:** $\frac{2\cos s-\cos 2s-1}{s^2}$

2. Find the Fourier cosine transform of e^{-ax} **Ans:** $F\left(e^{-ax}\right)=\frac{a}{a^2+s^2}$

<u>Infinite Fourier sine transforms</u>

I. Find the Fourier sine transforms of the following functions.

1. $f(x)=\frac{e^{-ax}}{x}$, $a>0$ **Ans:** $\tan^{-1}\left(\frac{s}{a}\right)$

2. $f(x)=e^{-|x|}$, **Ans:** $F_s(s)=\frac{s}{1+s^2}$

3. $f(x)=\begin{cases}\sin x & 0<x<a\\ 0 & x\ge a\end{cases}$ **Ans:** $\frac{1}{2}\left[\frac{\sin(s-1)a}{s-1}-\frac{\sin(s+1)a}{s+1}\right]$

4. $f(x)=\begin{cases}4x & 0<x<1\\ 4-x & 1<x<4\\ 0 & x>4\end{cases}$ **Ans:** $\left(\frac{5\sin s-\sin 4s}{s^2}\right)-\left(\frac{\cos s}{s}\right)$

5. Find the Fourier sine transform of e^{-ax} , $x\ge 0,\ a\ge 0$ **Ans:** $F_s(e^{-ax})=\frac{s}{a^2+s^2}$

<u>Infinite Inverse Fourier cosine transforms</u>

Find the inverse Fourier cosine transforms of the following functions

1. $F_c(s)=\begin{cases}\frac{1}{2\pi}\left(a-\frac{s}{2}\right) & 0<s<2a\\ 0 & s\geq 2a\end{cases}$, **Ans:** $f(x)=\frac{1}{\pi^2}\frac{\sin^2(ax)}{x^2}$

2. Solve the integral equation $\int_0^\infty f(\theta)\, cos\alpha\theta\, d\theta=\begin{cases}1-\alpha, & 0\leq\alpha\leq 1\\ 0, & \alpha>1\end{cases}$. Hence evaluate $\int_0^\infty \frac{\sin^2 t}{t^2}\, dt$.

3. Solve the integral equation $\int_0^\infty f(x)\, cos\alpha\, dx=e^{-\alpha}$. Hence show that $\int_0^\infty \frac{\cos\alpha x}{1+x^2}\, dx=\frac{\pi}{2}e^{-\alpha}$.

<u>Infinite Inverse Fourier sine transforms</u>

1. Find the inverse Fourier sine transforms of the following functions

$\frac{e^{-as}}{s}, a>0$ **Ans:** $\frac{2}{\pi}\tan^{-1}\left(\frac{x}{a}\right)$

2. Solve the integral equation $\int_0^\infty f(x)\, sintx\, dx=\begin{cases}1, & 0\leq t\leq 1\\ 2, & 1\leq t\leq 2.\\ 0, & t\geq 2\end{cases}$

3. Solve the integral equation $\int_0^\infty f(\theta)\, sin\alpha\theta\, d\theta=\begin{cases}1-\alpha, & 0\leq\alpha\leq 1\\ 0, & \alpha>1\end{cases}$.

<u>Convolution</u>

The convolution of two functions $f(x)$ and $g(x)$ over the interval $(-\infty,\infty)$ is defined as

$$f*g=\int_{-\infty}^{\infty} f(u)\,g(x-u)\,du=h(x)$$

<u>Convolution theorem</u>

Statement: The Fourier transform of the convolution of $f(x)$ and $g(x)$ is the product of their Fourier transforms.

i.e., $F\{f(x)*g(x)\}=F\{f(x)\}\cdot F\{g(x)\}$

Problems

1. Use convolution theorem to find the inverse Fourier transform of $\frac{i}{(1+s^2)^2}$, given that $\frac{2}{(1+s^2)}$ is the Fourier transform of $e^{-|x|}$.

2. By employing the convolution theorem, show that inverse Fourier transform of $e^{-\frac{s^2}{2}}$ is $\frac{1}{\sqrt{2\pi}}e^{-x^2/2}$

3. Using convolution theorem, show that $F^{-1}\left[\frac{1}{(a^2+s^2)(b^2+s^2)}\right]=\frac{1}{4ab}\int_{-\infty}^{\infty} e^{-(a|u|+b|x-u|)}du$